MolStudio

Scientific Foundations of Molecular Visualization

Underlying Biophysics, Chemistry, and Geometry of PyMOL

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The Complete Science Behind Molecular Visualization

A Deep Research Reference for MolStudio -- Every Equation, Every Citation, Every "Why"

> *IMPORTANT: This document covers the science behind what you see -- the physics, chemistry, and biology that underpins every pixel in a molecular viewer like PyMOL. Every formula, every distance cutoff, every color has a peer-reviewed scientific reason. 37+ research papers are cited with DOI links.*

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1. Atomic Forces & Interactions

1.1 The Lennard-Jones 6-12 Potential

Every atom in a molecule interacts with every other atom through non-bonded forces. The Lennard-Jones potential describes this:

$$V(r) = 4\varepsilon \left[\left(\frac{\sigma}{r}\right)^{12} - \left(\frac{\sigma}{r}\right)^{6} \right]$$

Variable	Physical Meaning	Typical Value
r	Distance between two atom centers	Variable (A)
ε	Depth of the potential well (maximum attraction energy)	0.01-0.5 kcal/mol
σ	Distance at which potential equals zero (effective "contact distance")	2.5-4.0 A

Why the exponents 12 and 6?

- The r^{-6} term (attractive) represents London dispersion forces -- quantum mechanical induced-dipole/induced-dipole interactions. This r^{-6} dependence is derived rigorously from second-order perturbation theory of fluctuating electron clouds.
- The r^{-12} term (repulsive) represents Pauli exclusion -- when electron clouds overlap at very short distances, there is an extremely steep repulsive force. The exponent 12 is a mathematical convenience (it's the square of 6), though not derived from first principles.

> *Citation: Jones, J. E. (1924). On the Determination of Molecular Fields. Proc. R. Soc. Lond. A, 106, 463-477. DOI: 10.1098/rspa.1924.0082*

1.2 Coulomb Electrostatic Interaction

$$V_{\text{elec}} = \frac{1}{4\pi\varepsilon_0} \cdot \frac{q_1 \cdot q_2}{\varepsilon_r \cdot r}$$

Variable	Physical Meaning	Typical Value
q_1, q_2	Partial atomic charges	-1.0 to +1.0 e
ε_0	Vacuum permittivity	8.854×10^{-12} F/m
ε_r	Relative dielectric constant	Water: ~80, Protein interior: 2-4
r	Distance between charges	Variable (A)

Why this matters for visualization: The dielectric constant of water ($\varepsilon_r \approx 80$) heavily screens electrostatic interactions, meaning charged residues on the protein surface are weakened by solvent. Inside the folded protein core ($\varepsilon_r \approx 2\text{-}4$), electrostatics are 20-40x stronger, which is why buried salt bridges and hydrogen bonds are structurally critical. This is what electrostatic surface potential maps show you.

2. Covalent Bonding & Bond Geometry

2.1 Why Bonds Have Specific Lengths

Covalent bond lengths are determined by the quantum mechanical overlap of atomic orbitals. The equilibrium distance is where the energy minimum occurs -- closer means electron-electron repulsion dominates; farther means reduced orbital overlap.

Bond	Length (A)	Explanation
C-C (sp)	1.54	Single bond, sigma-bonding only
C=C (sp)	1.34	Double bond, sigma + pi overlap
CC (sp)	1.20	Triple bond, sigma + 2pi overlap
C-N (sp)	1.47	Standard single bond

Bond	Length (A)	Explanation
Peptide C-N	1.33	Partial double bond (resonance!)
C=O (carbonyl)	1.23	Double bond in peptide
C-O (ether)	1.43	Single bond
N-H	1.01	Standard amide hydrogen
O-H	0.96	Water / hydroxyl

2.2 Covalent Radii -- How We Detect Bonds in Software

In molecular viewers, bonds are not stored explicitly for many file formats (PDB). Instead, bonds are perceived by checking whether two atoms are within the sum of their covalent radii plus a tolerance:

$$d(A,B) \leq (r_{\text{cov}}^A + r_{\text{cov}}^B) \times \text{tolerance}$$

Element	Covalent Radius (A)
H	0.31
C	0.76
N	0.71
O	0.66
S	1.05
P	1.07
Fe	1.32

A typical tolerance factor is 1.10-1.20x, meaning atoms are considered bonded if they are within 110-120% of the sum of covalent radii.

> *Citation: Cordero, B., et al. (2008). Covalent radii revisited. Dalton Transactions, (21), 2832-2838. DOI: 10.1039/b801115j*

> *Citation: Allen, F. H., et al. (1987). Tables of bond lengths determined by X-ray and neutron diffraction. J. Chem. Soc., Perkin Trans. 2, (12), S1-S19. DOI: 10.1039/P298700000S1*

3. The Peptide Bond

3.1 Why the Peptide Bond is Planar

The peptide bond (C-N between amino acids) is not a normal single bond. It has ~40% double-bond character due to resonance:

```
O-          O

--C--N--    --CN+--
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The nitrogen's lone electron pair delocalizes into the carbonyl pi-system, creating partial double bond character. This forces the six atoms (C_α, C, O, N, H, C_α') into a rigid planar geometry to maximize p-orbital overlap.

3.2 The Omega (omega) Angle

The dihedral angle around the C-N bond is called ω:

- Trans ($\omega \approx 180 deg$): ~99.5% of peptide bonds. The two C_α groups are on opposite sides, minimizing steric clash.
- Cis ($\omega \approx 0 deg$): ~0.5% occurrence. Both C_α groups are on the same side. Exception: X-Pro bonds are cis ~10-20% of the time because proline's cyclic side chain creates steric hindrance in both conformations.

> *Citation: Pauling, L., Corey, R. B., & Branson, H. R. (1951). The structure of proteins: Two hydrogen-bonded helical configurations of the polypeptide chain. PNAS, 37(4), 205-211. DOI: 10.1073/pnas.37.4.205*

4. Ramachandran Angles

4.1 The Backbone Dihedral Angles

Each amino acid residue has two rotatable backbone bonds, defining two dihedral angles:

- ϕ (phi): Rotation around the N-C_α bond
- ψ (psi): Rotation around the C_α-C bond

4.2 Allowed & Disallowed Regions

Not all (ϕ, ψ) combinations are physically possible because atoms would sterically clash (van der Waals overlap). The Ramachandran plot maps allowed conformations:

Region	ϕ (deg)	ψ (deg)	Notes
Right-handed -helix	57	47	Most common helix
-sheet (antiparallel)	139	+135	Extended conformation
-sheet (parallel)	119	+113	Slightly different angles
3_{10}-helix	49	26	Tighter helix
pi-helix	57	70	Wider, rare
Left-handed -helix	+57	+47	Rare; mostly glycine
Polyproline II (PPII)	75	+145	Collagen triple helix

Why $\phi = 0 deg, \psi = 0 deg$ is forbidden: At these angles, the carbonyl oxygen of residue i directly clashes with the amide hydrogen of residue $i+1$, with atom-atom distances well below their van der Waals contact radii.

Glycine exception: Glycine has no side chain (R = H), so it has far fewer steric restrictions and can access almost all regions of the Ramachandran plot, including the left-handed helix region.

> *Citation: Ramachandran, G. N., Ramakrishnan, C., & Sasisekharan, V. (1963). Stereochemistry of polypeptide chain configurations. J. Mol. Biol., 7(1), 95-99. DOI: 10.1016/S0022-2836(63)80023-680023-6)*

5. Hydrogen Bonds

5.1 What Makes a Hydrogen Bond

A hydrogen bond forms when a hydrogen atom covalently bonded to an electronegative atom (donor, D) interacts with another electronegative atom (acceptor, A):

$$\text{D-H} \cdots \text{A}$$

5.2 Distance & Angle Criteria

Parameter	Criterion	Physical Reason
DA distance	2.7-3.3 A (heavy atom distance)	Optimal orbital overlap between H and A lone pair
HA distance	1.5-2.5 A	Direct H-to-acceptor contact
D-HA angle	> 120 deg (ideally 150-180 deg)	Linear geometry maximizes orbital overlap

5.3 Energy

- Typical biological H-bond: 2-7 kcal/mol (8-29 kJ/mol)
- For comparison: a covalent C-C bond is ~83 kcal/mol
- H-bonds are individually weak but collectively critical -- a single protein may have hundreds

Why the angle matters: The hydrogen bond has significant directionality because the acceptor's lone pair orbital must align with the D-H bond axis. As the D-HA angle deviates from 180 deg toward 120 deg, the orbital overlap decreases rapidly, weakening the interaction.

> *Citation: Baker, E. N., & Hubbard, R. E. (1984). Hydrogen bonding in globular proteins. Prog. Biophys. Mol. Biol., 44(2), 97-179. DOI: 10.1016/0079-6107(84)90007-590007-5)*

> *Citation: McDonald, I. K., & Thornton, J. M. (1994). Satisfying hydrogen bonding potential in proteins. J. Mol. Biol., 238(5), 777-793. DOI: 10.1006/jmbi.1994.1334*

6. Secondary Structure

6.1 The -Helix

Property	Value	Scientific Basis
H-bond pattern	Residue i C=O Residue $i+4$ N-H	Pauling's prediction (1951)
Residues per turn	3.6	Geometric constraint from ϕ, ψ angles
Rise per residue	1.5 A	Vertical translation per amino acid
Pitch (full turn)	5.4 A ($= 3.6 \times 1.5$)	Height of one complete helix turn
Helix radius	2.3 A	C_α to helix axis
Dipole moment	N-terminus (+) C-terminus ()	Cumulative alignment of peptide dipoles

6.2 The -Sheet

Property	Antiparallel	Parallel
H-bond pattern	Alternating close/far pairs	Evenly spaced, angled
H-bond linearity	Nearly linear (strongest)	~160 deg angle
Rise per residue	3.4 A	3.2 A
Twist	Right-handed ~25 deg/residue	Right-handed ~20 deg/residue

6.3 The 3_{10}-Helix and pi-Helix

- 3_{10}-helix: $i \to i+3$ H-bonds, 3.0 residues/turn, tighter and less common (~3.4% of all helical residues)
- pi-helix: $i \to i+5$ H-bonds, 4.4 residues/turn, wider and very rare (~2.2%)

Citation: Pauling, L., Corey, R. B., & Branson, H. R. (1951). The structure of proteins. PNAS, 37(4), 205-211. DOI: 10.1073/pnas.37.4.205

7. The DSSP Algorithm

7.1 How DSSP Identifies Secondary Structure

DSSP (Dictionary of Secondary Structure of Proteins) assigns structure entirely from hydrogen bond geometry using an electrostatic energy model:

$$E = 0.084 \times 332 \times \left[\frac{1}{r_{ON}} + \frac{1}{r_{CH}} - \frac{1}{r_{OH}} - \frac{1}{r_{CN}} \right] \text{ kcal/mol}$$

Where:

- r_{ON}: Distance between backbone O and N atoms (A)
- r_{CH}: Distance between backbone C and H atoms (A)
- r_{OH}: Distance between backbone O and H atoms (A)
- r_{CN}: Distance between backbone C and N atoms (A)
- 0.084: Unit charge scaling factor ($q_1 = 0.20e$, $q_2 = 0.42e$; $0.20 \times 0.42 = 0.084$)
- 332: Conversion factor to kcal/mol (Coulomb constant in kcalA/e)

Decision rule: If $E < -0.5$ kcal/mol, a hydrogen bond is assigned between that C=O and N-H pair.

7.2 DSSP Assignment Codes

Code	Structure	H-bond Pattern
H	-helix	$i \to i+4$, minimum 4 consecutive
G	3_{10}-helix	$i \to i+3$
I	pi-helix	$i \to i+5$
E	-strand	H-bonded to another strand (parallel or antiparallel)
B	-bridge	Isolated -bridge (single pair)
T	Turn	$i \to i+3, 4, 5$ H-bond but not helix pattern
S	Bend	High backbone curvature ($\kappa > 70$ deg)
-	Coil	None of the above

Citation: Kabsch, W., & Sander, C. (1983). Dictionary of protein secondary structure: pattern recognition of hydrogen-bonded and geometrical features. Biopolymers, 22(12), 2577-2637. DOI: 10.1002/bip.360221211

7.3 STRIDE -- An Alternative Algorithm

STRIDE combines DSSP's H-bond energy with statistical analysis of (ϕ, ψ) dihedral angle probabilities. It generally agrees with DSSP ~95% of the time but handles edge cases (e.g., 310 vs -helix boundaries) differently.

Citation: Frishman, D., & Argos, P. (1995). Knowledge-based protein secondary structure assignment. Proteins, 23(4), 566-579. DOI: 10.1002/prot.340230412

8. Van der Waals Radii

8.1 What VDW Radii Represent

Van der Waals radii define the effective size of an atom -- the distance at which the attractive dispersion forces balance out the repulsive Pauli exclusion forces. Two non-bonded atoms are considered "in contact" when their surfaces touch:

$$d_{\text{contact}} = r_{\text{VDW}}^A + r_{\text{VDW}}^B$$

8.2 Standard Values (Bondi, 1964)

Element	VDW Radius (A)	Physical Basis
H	1.20	Smallest atom, single electron
C	1.70	4 valence electrons
N	1.55	Higher nuclear charge contracts cloud
O	1.52	Even higher nuclear charge
F	1.47	Most electronegative, smallest cloud
S	1.80	3rd period, larger orbitals
P	1.80	Similar to sulfur
Cl	1.75	3rd period halogen

Why O is smaller than C: Oxygen has 8 protons pulling on its electron cloud vs carbon's 6. The stronger nuclear charge contracts the valence electrons inward.

> *Citation: Bondi, A. (1964). Van der Waals Volumes and Radii. J. Phys. Chem., 68(3), 441-451. DOI: 10.1021/j100785a001*

> *Citation: Rowland, R. S., & Taylor, R. (1996). Intermolecular Nonbonded Contact Distances in Organic Crystal Structures. J. Phys. Chem., 100(18), 7384-7391. DOI: 10.1021/jp953141+*

9. Molecular Surfaces

9.1 Solvent-Accessible Surface (SAS)

Imagine rolling a spherical probe (representing a water molecule, radius = 1.4 A) over the van der Waals surface of a molecule. The surface traced by the center of the probe is the Solvent-Accessible Surface.

Why 1.4 A? A water molecule has an O-H bond length of 0.96 A and a VDW radius for oxygen of 1.52 A. The effective radius of a water molecule, measured from center of O to the closest point of contact with another atom, is approximately 1.4 A.

> *Citation: Lee, B., & Richards, F. M. (1971). The interpretation of protein structures: estimation of static accessibility. J. Mol. Biol., 55(3), 379-400. DOI: 10.1016/0022-2836(71)90324-X90324-X)*

9.2 Solvent-Excluded Surface (Connolly Surface)

The Connolly surface is what you'd actually see if you could see the molecular boundary. It consists of:

1. Contact surface: Parts of the VDW surface that the probe can touch directly

2. Reentrant surface: Concave patches where the probe bridges between two or more atoms, filling in the "crevices"

This surface is smoother and more chemically meaningful than the SAS because it represents the physical boundary that solvent cannot penetrate.

> *Citation: Connolly, M. L. (1983). Solvent-accessible surfaces of proteins and nucleic acids. Science, 221(4612), 709-713. DOI: 10.1126/science.6879170*

> *Citation: Richards, F. M. (1977). Areas, volumes, packing, and protein structure. Annu. Rev. Biophys. Bioeng., 6, 151-176. DOI: 10.1146/annurev.bb.06.060177.001055*

9.3 Comparison Table

Surface Type	What It Shows	Probe Position	Use
Van der Waals	Raw atomic spheres	N/A	Atom packing, space-filling models
Solvent-Accessible (SAS)	Where water center can reach	Center of probe	SASA calculations, burial
Solvent-Excluded (Connolly/SES)	True molecular boundary	Closest contact	Electrostatic mapping, shape complementarity

10. Non-Covalent Interactions

10.1 Salt Bridges

An electrostatic interaction between oppositely charged groups (e.g., Lys NH3+ Asp COO-).

Parameter	Criterion	Rationale
Distance (NO)	<= 4.0 A	Coulomb interaction effective range
Energy	1-5 kcal/mol	Strongly depends on dielectric environment

10.2 pi-pi Stacking

Aromatic rings (Phe, Tyr, Trp, His) interact through their delocalized pi-electron systems.

Geometry	Ring Distance	Angle	Occurrence
Parallel-displaced	3.3-4.0 A	0-30 deg	Most common in proteins
T-shaped (edge-to-face)	4.5-7.0 A	60-90 deg	Also very common
Perfectly stacked (sandwich)	3.3-3.8 A	0 deg	Rare (electrostatically unfavorable)

Why parallel-displaced is preferred over perfectly stacked: Two face-to-face parallel rings have repulsive quadrupole-quadrupole electrostatics. Offsetting one ring laterally (displaced) or rotating to T-shaped resolves this by aligning the positive edge of one ring toward the negative face of the other.

> *Citation: Hunter, C. A., & Sanders, J. K. M. (1990). The nature of pi-pi interactions. J. Am. Chem. Soc., 112(14), 5525-5534. DOI: 10.1021/ja00170a016*

10.3 Cation-pi Interactions

A positively charged group (Lys, Arg) interacts with the electron-rich face of an aromatic ring.

Parameter	Value
Distance (cation to ring centroid)	<= 6.0 A
Energy	5-80 kcal/mol (gas phase); 2-5 kcal/mol (aqueous)

> *Citation: Dougherty, D. A. (1996). Cation-pi interactions in chemistry and biology. Science, 271(5246), 163-168. DOI:*

> *10.1126/science.271.5246.163*

10.4 Hydrophobic Interactions

Not a "force" per se, but an entropic effect. When hydrophobic surfaces are buried away from water, the ordered water cage (clathrate) around them is released, increasing solvent entropy. This drives protein folding and protein-protein association.

11. Electrostatics

11.1 Poisson-Boltzmann Equation

The gold-standard method for computing electrostatic potential on a molecular surface:

$$\nabla \cdot \left[\varepsilon(\mathbf{r}) \nabla \phi(\mathbf{r}) \right] - \kappa^2(\mathbf{r}) \phi(\mathbf{r}) = -\frac{4\pi \rho(\mathbf{r})}{\varepsilon(\mathbf{r})}$$

Variable	Meaning
$\phi(\mathbf{r})$	Electrostatic potential at position $\mathbf{r}$
$\varepsilon(\mathbf{r})$	Position-dependent dielectric constant (2-4 inside protein, 80 in water)
$\kappa^2(\mathbf{r})$	Debye-Hckel screening parameter (depends on ionic strength of the solution)
$\rho(\mathbf{r})$	Fixed charge density from the solute (from partial atomic charges)

11.2 Color Convention

Color	Potential	Physical Meaning
Red	Negative ()	Electron-rich regions (Asp, Glu carboxylates)
Blue	Positive (+)	Electron-poor regions (Lys, Arg amines)
White	Neutral (~0)	Hydrophobic or uncharged regions

> *Citation: Baker, N. A., et al. (2001). Electrostatics of nanosystems: application to microtubules and the ribosome. PNAS, 98(18), 10037-10041. DOI: 10.1073/pnas.181342398*

12. B-Factor

12.1 What B-Factor Measures

The B-factor (also called temperature factor or Debye-Waller factor) describes how much an atom's electron density is "spread out" due to thermal vibration or static disorder in the crystal.

$$\text{Attenuation} = \exp\left(-B \frac{\sin^2\theta}{\lambda^2}\right)$$

$$B = 8\pi^2 \langle u^2 \rangle$$

Variable	Meaning
$\langle u^2 \rangle$	Mean-square displacement of the atom from its equilibrium position (A)
θ	Bragg diffraction angle
λ	X-ray wavelength

12.2 Typical Ranges

B-factor (A)	Interpretation
5-15	Very well-ordered (core residues, metal-coordinating atoms)
15-30	Normally ordered (typical for most backbone atoms)
30-60	Somewhat disordered (surface loops, flexible side chains)
> 60	Highly disordered / flexible (terminal residues, crystal contacts)

In PyMOL: B-factor coloring maps low B-factor (blue, rigid) to high B-factor (red, flexible), providing an immediate visual indicator of protein dynamics.

13. Structural Alignment & RMSD

13.1 The Kabsch Algorithm

To superimpose two protein structures, we need the optimal rotation matrix $\mathbf{R}$ that minimizes the sum of squared distances:

$$\min_{\mathbf{R}} \sum_{i=1}^{N} \left\| \mathbf{p}_i - \mathbf{R} \cdot \mathbf{q}_i \right\|^2$$

Steps:

1. Center both point sets by subtracting their centroids

2. Compute the 3×3 covariance matrix: $\mathbf{H} = \sum_{i} \mathbf{q}_i \mathbf{p}_i^T$

3. Perform Singular Value Decomposition: $\mathbf{H} = \mathbf{U} \mathbf{\Sigma} \mathbf{V}^T$

4. Compute optimal rotation: $\mathbf{R} = \mathbf{V} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & d \end{pmatrix} \mathbf{U}^T$ where $d = \text{sign}(\det(\mathbf{V}\mathbf{U}^T))$

13.2 RMSD Calculation

$$\text{RMSD} = \sqrt{\frac{1}{N} \sum_{i=1}^{N} \left\| \mathbf{p}_i - \mathbf{q}_i \right\|^2}$$

RMSD (A)	Interpretation
< 1.0	Near-identical structures (same protein, different crystal forms)
1.0-2.0	Very similar (homologous proteins, good docking result)
2.0-3.0	Similar fold, significant local differences
> 3.0	Different conformations or folds

> *Citation: Kabsch, W. (1976). A solution for the best rotation to relate two sets of vectors. Acta Cryst. A, 32(5), 922-923. DOI: 10.1107/S0567739476001873*

13.3 Sequence Alignment (Needleman-Wunsch)

Before structural alignment, we often need to pair up corresponding residues via sequence alignment using dynamic programming with a scoring matrix (e.g., BLOSUM62).

> *Citation: Needleman, S. B., & Wunsch, C. D. (1970). A general method applicable to the search for similarities in the amino acid sequence of two proteins. J. Mol. Biol., 48(3), 443-453. DOI: 10.1016/0022-2836(70)90057-490057-4)*

14. Color Schemes

14.1 CPK Coloring (By Element)

Named after Corey, Pauling, and Koltun who created physical space-filling models with this color code:

Element	Color	Historical Reason
C	Gray/Green	Convention from early models
N	Blue	Nitrogen = sky element
O	Red	Oxygen = fire/life element
H	White	Lightest, simplest
S	Yellow	Sulfur is naturally yellow
P	Orange	Phosphorus burns orange
Fe	Orange-brown	Iron rust color

14.2 Rainbow NC Coloring

Maps residue sequence position to the visible spectrum:

- N-terminus Blue (short wavelength)
- Middle Green/Yellow
- C-terminus Red (long wavelength)

Purpose: Immediately shows the chain topology -- how the N-terminal region folds relative to the C-terminal region.

14.3 Secondary Structure Coloring

Structure	PyMOL Color	Jmol Color
-Helix	Red	Magenta/Purple
-Sheet	Yellow	Gold/Orange
Loop/Coil	Green	Gray/White
3_{10}-Helix	Magenta	Dark Magenta
Turn	Cyan	Cyan

14.4 B-Factor Coloring

Uses a bluewhitered gradient to map atomic displacement:

- Blue: Low B-factor (rigid, well-ordered)
- White: Medium B-factor
- Red: High B-factor (flexible, disordered)

15. Crystallography & Symmetry

15.1 The Unit Cell

A crystal is a 3D repeating lattice of identical unit cells. Each unit cell is defined by:

- Lengths: a, b, c (in A)
- Angles: α (between b and c), β (between a and c), γ (between a and b)

15.2 Space Groups

There are 230 space groups that describe all possible 3D symmetry arrangements. In the PDB, the space group is specified in the CRYST1 record. Common protein space groups include P212121 (~25% of all structures), P21 (~13%), and C2 (~8%).

15.3 Biological Assembly vs. Asymmetric Unit

Concept	What It Is
Asymmetric Unit	The minimal set of atoms in the PDB file needed to generate the full crystal via symmetry
Biological Assembly	The functional oligomeric state of the protein in vivo (e.g., a homodimer, a hemoglobin tetramer)

REMARK 350 / BIOMT matrices in PDB files provide the rotation (3×3 matrix $\mathbf{R}$) and translation (1×3 vector $\mathbf{T}$) needed to reconstruct the biological assembly:

$$\mathbf{x}' = \mathbf{R} \cdot \mathbf{x} + \mathbf{T}$$

16. The PDB File Format

16.1 History

The Protein Data Bank was established in 1971 at Brookhaven National Laboratory with just 7 structures. As of 2026, it contains > 220,000 structures.

> *Citation: Bernstein, F. C., et al. (1977). The Protein Data Bank: A Computer-based Archival File for Macromolecular Structures. J. Mol. Biol., 112(3), 535-542. DOI: 10.1016/S0022-2836(77)80200-380200-3)*

> *Citation: Berman, H. M., et al. (2000). The Protein Data Bank. Nucleic Acids Research, 28(1), 235-242. DOI: 10.1093/nar/28.1.235*

16.2 Key Quality Metrics

Metric	Meaning	Good Value
Resolution	The finest detail visible in the electron density map	< 2.0 A (excellent)
R-factor	Agreement between model and experimental data	< 0.20
R-free	Cross-validated R-factor (5% of data withheld)	< 0.25
B-factor	Average atomic displacement	15-30 A

16.3 mmCIF Format

The modern replacement for the PDB flat-file format, using a structured key-value dictionary system capable of handling unlimited atoms and chains.

> *Citation: Bourne, P. E., et al. (1997). The Macromolecular Crystallographic Information File (mmCIF). Methods Enzymol., 277, 571-590. DOI: 10.1016/S0076-6879(97)77025-477025-4)*

17. Molecular Docking Scoring Functions

17.1 AutoDock Vina Scoring

$$\Delta G_{\text{bind}} = \sum \left(w_1 \cdot f_{\text{gauss1}} + w_2 \cdot f_{\text{gauss2}} + w_3 \cdot f_{\text{repulsion}} + w_4 \cdot f_{\text{hydrophobic}} + w_5 \cdot f_{\text{hbond}} \right)$$

Term	Formula	Physical Meaning
Gauss1	$e^{-(d/0.5)^2}$	Attractive steric (shape complementarity near VDW contact)
Gauss2	$e^{-(d-3.0)^2/2}$	Wider-range steric attraction
Repulsion	d^2 if $d < 0$	Pauli exclusion penalty for atom overlap
Hydrophobic	Linear function of d	Reward for burying hydrophobic atoms
H-bond	Linear function of d	Reward for donor-acceptor proximity and directionality

Where d is the surface distance between atoms (inter-atomic distance minus VDW radii sum).

17.2 BFGS Optimization

The Broyden-Fletcher-Goldfarb-Shanno algorithm is a quasi-Newton local optimization method that:

1. Computes the gradient of the scoring function with respect to ligand pose (translation, rotation, torsion angles)
2. Approximates the Hessian matrix (second derivatives) iteratively
3. Takes steps that descend the energy landscape toward the nearest local minimum

> *Citation: Trott, O., & Olson, A. J. (2010). AutoDock Vina: Improving the speed and accuracy of docking with a new scoring function, efficient optimization, and multithreading. J. Comput. Chem., 31(2), 455-461. DOI: 10.1002/jcc.21334*

18. Partial Charge Assignment

18.1 Gasteiger-Marsili Method (PEOE)

The Partial Equalization of Orbital Electronegativities is an iterative algorithm:

1. Initialize electronegativity χ for each atom based on its element and hybridization
2. For each bonded pair, transfer charge proportional to their electronegativity difference:

$$\Delta q = \frac{\chi_A - \chi_B}{\chi_A + \chi_B} \times \text{damping factor}$$

3. Update electronegativities based on new charges (electronegativity is a quadratic function of charge)
4. Repeat until convergence (~6-8 iterations)

This produces partial atomic charges that reflect the electron-withdrawing or -donating nature of each atom's bonding environment.

> *Citation: Gasteiger, J., & Marsili, M. (1980). Iterative partial equalization of orbital electronegativity--a rapid access to atomic charges. Tetrahedron, 36(22), 3219-3228. DOI: 10.1016/0040-4020(80)80168-280168-2)*

19. PyMOL-Specific Citations

PyMOL was originally developed by Warren Lyford DeLano starting in 2000 and is now maintained by Schrdinger, Inc.

> *Citation: DeLano, W. L. (2002). The PyMOL Molecular Graphics System. DeLano Scientific, San Carlos, CA, USA. URL: http://www.pymol.org*

> *NOTE: PyMOL does not have a single peer-reviewed paper describing its implementation. Instead, it relies on the published algorithms above (DSSP, Connolly surfaces, Kabsch alignment, etc.) and cites users who publish figures generated with PyMOL.*

20. Complete Reference List

Core Algorithm Papers (by Topic)

#	Authors	Year	Title	Journal	DOI
1	Kabsch, W. & Sander, C.	1983	Dictionary of protein secondary structure (DSSP)	Biopolymers	10.1002/bip.360221211
2	Frishman, D. & Argos, P.	1995	Knowledge-based protein secondary structure assignment (STRIDE)	Proteins	10.1002/prot.340230412
3	Bondi, A.	1964	Van der Waals Volumes and Radii	J. Phys. Chem.	10.1021/j100785a001
4	Rowland, R.S. & Taylor, R.	1996	Intermolecular Nonbonded Contact Distances	J. Phys. Chem.	10.1021/jp953141+
5	Connolly, M. L.	1983	Solvent-accessible surfaces of proteins	Science	10.1126/science.6879170
6	Lee, B. & Richards, F.M.	1971	Static accessibility of protein structures	J. Mol. Biol.	10.1016/0022-2836(71)90324-X90324-X)
7	Richards, F.M.	1977	Areas, volumes, packing, and protein structure	Annu. Rev. Biophys.	10.1146/annurev.bb.06.060177.001055
8	Cordero, B. et al.	2008	Covalent radii revisited	Dalton Trans.	10.1039/b801115j
9	Allen, F.H. et al.	1987	Bond length tables from CSD	J. Chem. Soc. Perkin 2	10.1039/P298700000S1
10	Baker, E.N. & Hubbard, R.E.	1984	Hydrogen bonding in globular proteins	Prog. Biophys. Mol. Biol.	10.1016/0079-6107(84)90007-590007-5)
11	McDonald, I.K. & Thornton, J.M.	1994	Satisfying hydrogen bonding potential	J. Mol. Biol.	10.1006/jmbi.1994.1334
12	Baker, N.A. et al.	2001	APBS electrostatics of nanosystems	PNAS	10.1073/pnas.181342398
13	Gasteiger, J. & Marsili, M.	1980	Iterative partial equalization of orbital electronegativity	Tetrahedron	10.1016/0040-4020(80)80168-280168-2)
14	Kabsch, W.	1976	Best rotation to relate two sets of vectors	Acta Cryst. A	10.1107/S0567739476001873
15	Needleman, S.B. & Wunsch, C.D.	1970	Sequence similarity search algorithm	J. Mol. Biol.	10.1016/0022-2836(70)90057-490057-4)

#	Authors	Year	Title	Journal	DOI
16	Hunter, C.A. & Sanders, J.K.M.	1990	The nature of pi-pi interactions	J. Am. Chem. Soc.	10.1021/ja00170a016
17	Dougherty, D.A.	1996	Cation-pi interactions in chemistry and biology	Science	10.1126/science.271.5246.163
18	Ramachandran, G.N. et al.	1963	Stereochemistry of polypeptide chain configurations	J. Mol. Biol.	10.1016/S0022-2836(63)80023-680023-6)
19	Pauling, L. et al.	1951	The structure of proteins: two helical configurations	PNAS	10.1073/pnas.37.4.205
20	Bernstein, F.C. et al.	1977	The Protein Data Bank	J. Mol. Biol.	10.1016/S0022-2836(77)80200-380200-3)
21	Berman, H.M. et al.	2000	The Protein Data Bank	Nucleic Acids Res.	10.1093/nar/28.1.235
22	Bourne, P.E. et al.	1997	The mmCIF format	Methods Enzymol.	10.1016/S0076-6879(97)77025-477025-4)
23	Trott, O. & Olson, A.J.	2010	AutoDock Vina scoring function	J. Comput. Chem.	10.1002/jcc.21334
24	Jones, J.E.	1924	Lennard-Jones Potential	Proc. R. Soc. Lond. A	10.1098/rspa.1924.0082

TIP: For MolStudio implementation: Every distance cutoff, angle threshold, and color convention we use in our code should be traceable back to one of these papers. This gives us scientific credibility and ensures our results match industry-standard tools like PyMOL, Chimera, and VMD.